

Troubleshooting Tools — Guided Notes ANSWER KEY

Unit 5.1.2 | CompTIA Network+ N10-009 Obj. 5.1 | N10-008 5.1

For instructor use / distribute after students complete the notes.

Question 1

The `ping` command sends _____ Echo Request packets and listens for _____ replies. A 'Request timed out' means no reply came back, while 'Destination host unreachable' means _____ explicitly reported the host is not reachable.

Answer: ICMP; Echo; a router along the path.

Question 2

In ping output, the TTL value tells you approximately how many _____ the packet traversed. If you ping a Linux host and receive TTL=56, approximately _____ hops separated you from the destination.

Answer: Hops (routers); 8 hops (Linux starts TTL at 64; $64 - 56 = 8$).

Question 3

`tracert`/`tracert` maps the Layer 3 path by sending packets with incrementing _____ values. When a packet's TTL reaches zero, the router sends back a _____ message revealing its IP address.

Answer: TTL (Time to Live); ICMP Time Exceeded.

Question 4

You run `nslookup portal.school.edu` and the Server line shows `10.1.1.5`. You then run `nslookup portal.school.edu 8.8.8.8` and get a different IP. What does this indicate, and what should you investigate?

Answer: The client's configured DNS server (10.1.1.5) is returning an incorrect record for that hostname. This could be DNS poisoning, a misconfigured zone file, or cache corruption. Investigate the DNS server's zone records and cache.

Question 5

A workstation running Windows shows a 169.254.x.x address in `ipconfig /all`. This is called _____ and means the client _____ to receive a DHCP lease. List two possible causes.

Answer: APIPA (Automatic Private IP Addressing); failed. Causes: DHCP server is down, no IP helper-address (relay) on the gateway, DHCP scope is exhausted, or the client is on the wrong VLAN with no DHCP server.

Question 6

`netstat -an` shows the state LISTENING on port 443 and ESTABLISHED on port 3306. What do these states tell you, and what two services are most likely running?

Answer: LISTENING means a service is waiting for inbound connections on that port — port 443 means a web server (HTTPS) is running. ESTABLISHED means an active TCP session exists — port 3306 is MySQL, so the host has an active database connection.

Question 7

In Wireshark, _____ filters limit what is captured (set before capture begins), while _____ filters limit what is displayed from an existing capture. If you suspect ARP poisoning, what display filter and what pattern would you look for?

Answer: Capture filters; display filters. Filter: `arp`. Look for: gratuitous ARP replies (responses sent without a request) where multiple different MAC addresses are claiming the same IP — especially the gateway's IP.

Question 8 — Real World Application

REAL WORLD: You are given a fiber optic cable from a patch panel to an AP that has no link light. You clean the connectors, verify the SFP types match, and verify the AP is powered. The link still won't come up. Name the hardware tool you use next, what it measures, and what result would tell you the cable has a break and where.

Answer: OTDR (Optical Time-Domain Reflectometer). It sends light pulses down the fiber and measures reflections, calculating attenuation and the distance to each event. A sharp drop in the trace at a specific distance (e.g., at 23 meters) indicates a break or severe bend at that point in the cable run.